

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 7

(1) Research Program/Project/Study Title: Development and evaluation of Rice Flakes with Oyster Mushroom Instant Cereals"	
Proponent Name:	Dr. Ace D. Barredo
Academic Rank:	Associate Professor II
Co-Authors (with academic rank):	Ma. Consuelo Yap, EdD- Asso Prof V Mary Ann D. Cervas,MA-HE- Asso Prof I
Campus/Satellite College/Address:	CapSU Roxas City- Main Campus
Telephone/Cell Phone/Fax Number:	09994044417
E-mail Address:	ace07barredo@gmail.com

Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	Instant rice flakes with oyster mushroom instant rice flakes with malunggay porridge, and instant rice flakes with dilis champorado
Component Leader:	Ma. Consuelo Y. Yap, EdD and Mary Ann D. Cervas,MA-HE
Academic Rank:	Asso Prof. V and Asso Prof. I

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
--	--

(4) Site of implementation							
Municipality/City:	Roxas City						
Province:	Capiz						
Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 7

(7) Sector/Commodity

Agriculture/Fruit and Vegetables

(8) Discipline/Field of Specialization

Food Development

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The need for instant and ready-to-eat foods has now become a necessity due to the fast-paced scenario where everybody wants things quick, easy, and accessible without compromising their quality and health benefits. The development of rice as a source of energy into value-added products that are ready in an instant is a response to the demands and also provides provision of emergency food to survivors of disasters.

This study would like to ascertain the acceptability of the prepared ready-to-eat meals which can be eaten without too much preparation. Locally available ingredients were utilized to give variation to the food products. One of these value-added products is the instant rice flakes with oyster mushrooms instant cereals. Furthermore, the shelf life of the rice flakes was observed in order to make sure of the duration of storage and expiration both at room and chilling temperatures.

A developmental-experimental method of research was employed in the preparation having three treatments and sensory evaluation for the acceptability using a nine-point Hedonic Scale checklist among 50 consumers based on qualities such as appearance, aroma, crunchiness, taste, and texture. Purposive sampling was utilized considering the working parents only who have busy schedules. Based on the survey mean results, Treatment B was "Liked Extremely" while Treatment A and C were both "Liked Very Much". Moreover, it took two months for the rice flakes to be fit for human consumption at room temperature but at chilling temperature still in good condition within 5 months of storage.

Thus, the acceptability of the rice flakes with oyster mushrooms instant cereals shows that busy working parents these days, do not eat snacks only to provide satisfaction for their hunger, but also to supply essential nutrients to the body. The supplementation of Oyster mushrooms in the rice flakes makes a great difference. The preparation of food and storage is essential for a longer shelf-life without compromising the food quality.

Keywords: Rice Flakes, Instant cereal, Oyster Mushrooms

(10) Introduction

Food habits of Filipino consumers are changing due to the increasing number of working women, development of modern lifestyle, rise in nuclear families, shortage of time, high literacy level among women, increase in consumer awareness, and demand for convenience goods. The number of working women is rising, thus there is no time to prepare food at home. Hence, this created the need for using ready-to-eat foods. In this fast-paced world, everybody wants things quick and easy. Food manufacturers have responded to these needs and over the years have developed foods that are ready in an instant.

Since the Philippines is also a disaster-prone country that requires preparedness for natural hazards such as the provision of emergency food to survivors of disasters, Briones as cited by Balibay (2020) said the Pack of Hope RTE Chicken Arroz Caldo is its first stage relief good that can address the immediate need for food for hungry disaster victims within 48 hours of the incident. The ready-to-eat meal can be consumed without preparation and is drinkable with a shelf life of at least 1 year, and can be transported on land, sea, and aerial drop. Likewise, DOST Region 2 Director Engr. Sancho Mabborang noted how the DOST Ready-To-Eat (RTE) Pack of Hope Chicken Arroz Caldo of the DOST-ITDI provides nutritious meals that do not require much preparation (Balibay, 2020).

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 3 of 7

The present study was the development and evaluation of rice flakes value-added products as such rice flakes with oyster mushrooms instant cereals. The acceptability of this value-added product was evaluated based on sensory qualities such as appearance, aroma, crunchiness, taste, and texture. Its storage quality was also determined in terms of temperature conditions.

This study generally aimed to utilize rice powder and oyster mushrooms as locally available food resources in instant cereals preparation.

1. Determine the acceptability of rice flakes with oyster mushrooms instant cereals in three different treatments based on the sensory qualities in terms of appearance, aroma, crunchiness, taste, and texture.
2. Find out the storage quality of rice flakes with oyster mushrooms instant cereals in terms of temperature.

The present study was based on the following prior arts such as:

Bender(2005) describes instant food products as dried foods that need water to get back to their original form such as tea, coffee, soup, precooked cereal product, etc. The amount of moisture removed from dehydrated food is commonly about 1-5%. In a nutshell, instant food products are food or drink that is dried, usually in the form of a powder, and can be prepared very quickly by adding hot water.

Rice flour is made from finely milled rice. Rice flour is nutritious and is extensively used to make fresh rice noodles and sweets. Rice flour is obtained from either brown rice or white rice. Rice flour has become increasingly popular as a raw material for extrusion cooking in the food industry. Rice flour is one of the foundation ingredients in gluten-free baking. It is considered a decent substitute for wheat flour, which often hampers the functioning of the digestive system.

The utilization of rice flour in various snack food products had been explored in the following studies:

Eight products of traditional/commercial importance were standardized incorporating Popular Low Grade Rice Varieties (PLRVs) as a major ingredient in the form of rice flour (burfi, noodles, and extruded snack products and vennaundalu (butter coated balls), palathalikalu (dough rolled into strips, steamed/cooked in milk); rice semolina (instant kheer mix and instant upma mix), and flaked rice (nutritious bar). The products were evaluated for nutritional, cooking quality characteristics, consumer acceptability, and shelf-life. Consumer acceptability of the PLR products was carried out with 60 farm women based on 9 points hedonic scale. Shelf-life of the products (packed in both metalized PP and PE pouches) was evaluated monthly for chemical, microbiological and sensory parameters. Utilization of PLR flakes in nutritious bars resulted in a highly crispy and crunchy texture, enhanced flavor, and taste as compared to the control. Extruded snacks (control and PLR) showed no microbial growth during the entire storage period. This study by Anitha and Rajyalakshmi (2014) signifies the utilitarian value of PLRVs in making products of convenience/commercial importance.

Likewise, in the study of Agbaje and etc. (2014), they formulated cereal bars using dry materials like glutinous rice, black Cummins, etc., and binding agents (honey and glucose syrup). The dried fruits were mixed with the dry ingredients and binding ingredients at different percentages. The cereal bars were assessed for mineral element contents using microwave-assisted digestion method. The consumer acceptability of the cereal bars was carried out using a 9 hedonic scale. It was concluded that the incorporation of halal/sunnah fruits into the production of cereal bars will still make it retain much of the nutritional and sensory properties.

There are 101 Oyster Mushroom Recipes, a collection by Sonoran Mushroom Company. It can be observed that most of these recipes utilize mushrooms as sauce, extender, and additive in different dishes, classic merienda favorites, soups, and snacks. It also includes processed foods like tocino,

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 7

bacon, and by-products. An example of a snack is the making of delicious and crunchy mushroom biscuits. These were prepared by using 300g of oyster mushroom powder and various ingredients (Wakchaure, Shirur, Manikandan, & Rana, 2010). These were evaluated for color, appearance, flavor, taste texture, and overall acceptability by using organoleptic evaluation on the nine-point Hedonic Scale through a panel of ten judges (Rangana, 1994). Likewise, mushrooms are a source of B vitamins (B2, B3, folate, B5), Phosphorus, Vitamin D, Selenium, Copper, and Potassium (Zhang, et al., 2016).

The research work of Shinde and Durgadevi (2017) investigated the effect of flavor addition on the quality of rice flakes with iron fortification by adding chocolate flavor to rice flakes as a result of coating. The chocolate sauce was coated with rice flakes using enrober. Chocolate-coated rice flakes were having high iron compounds than other variations. As the flakes were coated with cocoa powder, the Catechin content was also enriched in the flakes.

Synthesis:

The studies cited are the utilization of rice flour in baked products and rice flakes chocolate flavor which were used as baseline data in the development, formulation, and evaluation of the rice flakes value-added products in this study. Due to the high demand for a balanced instant food packed with important nutrients in just a few minutes of preparation, these breakfast cereals, utilized gluten-free powder rice flour made into flakes and are definitely healthy snack alternatives.

Moreover, the formulation and evaluation of the value-added products in the previous studies were used as a reference in the present study. However, modifications in the procedure in the preparation and utilization of localized food resources as ingredients were varied with an emphasis on the homemade taste of “Panlasang Pinoy”. The supplementation of Oyster mushrooms in the rice flakes would make a great difference.

(11)

The present study used the Nine Point Hedonic Scale scorecard as a research instrument. It looked into the five (5) sensory qualities such as appearance, aroma, crunchiness, taste, and texture as scoring variables in evaluating the product. Mean was used to determine the acceptability of the product based on the sensory qualities. There were three trials done to come up with the most acceptable rice flake value-added products in terms of sensory qualities. The finished product was evaluated with semi-trained panelists for the product formulation prior to consumers' acceptability of the value-added products of rice flakes among 50 working parents selected purposively.

Product Preparation of Rice Flakes with Oyster Mushrooms Instant Cereal

Ingredients: Rice flour, Fresh Oyster Mushroom (purée), Muscovado Sugar, Iodized salt, Vanilla extract, Water

Directions:

1. Preheat oven to 350 degrees. Line a baking sheet with parchment paper or foil and grease lightly with oil or baking spray.
2. Whisk together rice flour, muscovado sugar, salt, and fresh oyster mushroom. Add vanilla extract and water, a little at a time, stirring until the batter is smooth and thin.
3. Pour batter onto the prepared pan and spread out evenly about 1/4 - 1/3" thick.
4. Combine remaining rice flour with 1 teaspoon of water and mix until the mixture resembles coarse breadcrumbs.
5. Sprinkle the top of the batter in the pan with the rice crumbs.
6. Bake on the center rack for 10-15 minutes, keeping a close eye on it, until the dough has dried out and cracked.
7. Remove from oven and lower heat to 250. Let the pan cool, then use your hands to tear and crack the dough into small flakes. Return to oven and let bake on the center rack for about 45 minutes or until pieces are toasted, crisp, and golden. Let cool completely.
8. Pack the rice flakes together with dried fruits and nuts and sealed completely.
9. Store in an air-tight container in a cool dry place.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 5 of 7

(12) Results and Discussions

General Acceptability of Rice Flakes with Oyster Mushrooms Instant Cereals

Table 1 reflects the preferences of the 50 consumers as shown in the general acceptability of Rice Flakes with oyster mushrooms instant flakes in terms of appearance, aroma, crunchiness, taste, and texture. Generally, Treatment B was “Liked Extremely” and a potential product for development as shown by the grand mean ratings of 8.43. Treatments A and C were shown as “Liked Very Much” indicating a mean range of 7.90 – 7.93. However, consumers have generally preferred Treatment B among other proportions when preparing rice flakes with oyster mushrooms instant cereals. This result is parallel with the study of Wakchaure, et al (2010) in the making of delicious and crunchy mushroom biscuits. These were prepared by using 300g of oyster mushroom powder and various ingredients. Moreover, Rangana, (1994) used organoleptic evaluation on the nine-point Hedonic Scale for color, appearance, flavor, taste, texture, and overall acceptability through a panel of ten judges.

Table 1. General Acceptability of Rice Flakes (with Oyster Mushroom) as Evaluated by Consumers.

SENSORY QUALITY	Treatment A		Treatment B		Treatment C	
	Mean	QD	Mean	QD	Mean	QD
Appearance	7.88	LVM	8.42	LE	7.90	LVM
Aroma	8.02	LVM	8.42	LE	7.74	LVM
Taste	7.96	LVM	8.58	LE	7.88	LVM
Crunchiness	7.90	LVM	8.44	LE	7.88	LVM
Texture	7.88	LVM	8.30	LE	8.08	LVM
General Acceptability	7.93	LVM	8.43	LE	7.90	LVM

Legend:

Scale of Means

8.12 – 9.00

7.23 – 8.11

6.34 – 7.22

Qualitative Description

Liked Extremely

Liked Very Much

Liked Moderately

Storage Quality of Rice Flakes with Oyster Mushrooms Instant Cereal

Table 2 reveals the shelf-life of the sealed rice flakes with oyster mushrooms instant cereal when stored at room/normal temperature. Storage was made for two (2) months. The first and second months of storage showed no change in the product qualities. However, in its third month, the three (3) treatments were noticed to have undergone changes such as mold formation, unpleasant odor, and production of spots were observed.

Table 2 .Shelf-life of the Rice Flakes (with Oyster Mushroom) at RoomTemperature.

Variant	One-month Mold Formation, unpleasant odor	Two-months Mold Formation unpleasant odor	Three-months Mold Formation, unpleasant odor, production of spot
Treatment A	0	0	+

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 7

Treatment B	0	0	+
Treatment C	0	0	+

0 – absence of molds

+ – presence of molds

However, when stored at a chilling temperature for five (5) months, the rice flake with oyster mushroom instant cereal showed no change in the preparation during the first to fourth month of storage. But, during its five (5) months of storage, changes like mold, unpleasant odor, and production of spots were observed. Microbial proliferations in foods need certain conditions – namely available water (water activity), proper pH, right temperature and nutrients, and time. By controlling these conditions one can prevent microbial growth and extend the shelf life of food (IbeanuVN, 2015)

Table 3. Shelf-life of the Rice Flakes (with Oyster Mushroom) at Chilling Temperature.

Variant	One-month Mold Formation, unpleasant odor	Two-months Mold Formation, unpleasant odor	Five-months Mold Formation unpleasant odor, production of spot
Treatment A	0	0	+
Treatment B	0	0	+
Treatment C	0	0	+

(13) Conclusions and Recommendations

Conclusions

Based on the findings of the study, the following conclusions are drawn:

The acceptability of the rice flakes' with oyster mushroom instant cereals meets the demand of people with busy schedules and they still prefer nutritional and functional ready-to-eat food in an instant package with less preparation. The storage shelf life duration of rice flakes can be used for the provision of emergency food to victims and survivors of disasters. Likewise, supplementary food when there is a shortage of food supply.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are forwarded:

Packaging of the rice flakes' with oyster mushroom instant cereals would be approximately weighed for one serving only for convenience and affordability.

Increase the shelf life of rice flakes by improving the method of preparation using machines and technology in food preservation without compromising the quality and nutritional contents.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Rice flakes' with oyster mushroom instant cereals packed food products with "panlasang pinoy" attributes for product development and commercialization.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 7 of 7

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

References

- Agbaje, R., Hassan, C. Z., Arifin, N., & Rahman, A. A. (2014). Sensory preference and mineral contents of cereal bars made from glutinous rice flakes and Sunnah foods. *IOSR Journal of Environmental Science, Toxicology and Food Technology*, 8(12), 26-31.
- Anitha, G., & Rajyalakshmi, P. (2014). Value-added products with popular low-grade rice varieties of Andhra Pradesh. *Journal of food science and Technology*, 51(12), 3702-3711.
- Balibay, A. (2020) Emergency Ready-to-Eat food packs developed by DOST to be commercialized. Retrieved from <https://www.goodnewspilipinas.com/emergency-ready-to-eat-food-packs-developed-by-dost-to-be-commercialized/#:~:text=Briones%20said%2>
- Bender, D. A.(2005) "Instant foods." A Dictionary of Food and Nutrition. Encyclopedia.com
- IbeanuVN.,Ene-Obong, HN., Peter-Ogba GU. andOnyechi UA.,(2015). Microbiological evaluation and shelf life of seed flour mixes used for infant feeding in rural northern Nigeria. *African Journal of Biotechnology*. 14(20): 1718-1723.
- Rangana, S. (1994). Manual Analysis of fruits and vegetable production.
- Shinde, G., & Durgadevi, M. (2017). Development of Iron-Fortified Chocolate Flavoured Rice Flakes. *J Nutr Food Sci*, 8(649), 2.
- Wakchaure, G. C., Shirur, M., Manikandan, K., & Rana, L. (2010). Development and evaluation of oyster mushroom value-added products. *Mushroom Research*, 19(1), 40-44.
- Zhang JJ, Li Y, Zhou T, Xu DP, Zhang P, Li S, Li HB (2016) Bioactivities and health benefits of mushrooms mainly from China. *Molecules*. 21(7):938.